

Family-Conditioned Multiview Occupancy Fitting for Semantic Proxy Actors in a Synthetic UAV Digital-Twin Setting (SPPA-MVFit)

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Abstract

Novelty. We show that a *frozen semantic-family part graph* is a measurable 3D occupancy prior for lightweight multiview proxy fitting: under an input-matched, equal-budget comparison, family-conditioned SPPA-MVFit improves voxel occupancy over a nonsemantic eight-part graph when both fit the same calibrated top/side silhouettes. The optimizer is deliberately simple; the knowledge representation (family topology) is the object of inference—not a new neural reconstructor, not open-set word-to-3D, and not an Unreal systems demo as the scientific claim.

Method. SPPA-MVFit adjusts five shared parameters of a fixed eight-part primitive actor with a 31-candidate silhouette objective. Text-only and silhouette-fitted modes call the same actor builder; only θ changes.

Evidence. On a preregistered developer-held-out synthetic test ($n=240$ actors; six families; CSG-ID and implicit-ODD strata), the primary paired mean voxel-IoU difference SPPA-MVFit – Generic-MVFit is 0.190 (stratified bootstrap 95% CI [0.181, 0.199]), exceeding the prespecified +0.030 superiority margin. All twelve family $\times$ stratum cells are positive; OOD stratum mean $\Delta = 0.172$. Median fitter wall time is 9.4 ms (p95 10.6 ms) locally.

Boundary. Results are synthetic internal evidence only. Deployment context (descriptor/update contract, low-polygon runtime proxies) is secondary and is not used to claim photoreal image-to-3D SOTA, measured flight validity, or operator benefit.

Keywords: semantic 3D proxy; multiview silhouette fitting; family-conditioned part graph; occupancy IoU; synthetic spatial proxy; UAV digital-twin setting.

1 Introduction

Semantic-telemetry UAV digital twins replace part of the continuous video stream with object-level messages. A UAV-side perception stack can report object tracks, classes, confidence, positions, and approximate geometry. The ground station then has to represent those objects in an Unreal/Cesium digital twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and runtime loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes.

Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The scientific question is therefore not whether SPPA looks better than neural 3D generation. It should not. The measurable question—and the only confirmatory claim of this paper—is whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated silhouettes. Runtime persistence, low polygon count, and updateability are reported later as secondary engineering properties, not as the primary scientific result.

2 Contribution and Scope

The scientific contribution is **family-conditioned SPPA-MVFit**: evidence that a frozen semantic-family part graph improves 3D occupancy over an equal-budget nonsemantic eight-part graph under identical calibrated top/side silhouettes, parameter count, objective, bounds, initialization, and candidate budget. The knowledge representation under test is the family graph; the optimizer is deliberately simple and is not claimed as a novel fitting algorithm. The method is family-conditioned, not class-agnostic open-set fitting, and not universal image-to-3D reconstruction.

Secondary contributions provide deployment context only:

- **Shared generator contract.** Text-only and silhouette-fitted SPPA modes call the same actor builder and part graph; silhouette evidence changes only the five-parameter vector θ .
- **Runtime descriptor/update layer.** SPPA-DESC-0.2/ SPPA-UPD-0.2 separate create from pose/material/shape updates for optional Unreal consumption alongside curated assets.
- **Evidence boundary.** Confirmatory occupancy evidence is developer-held-out synthetic geometry. Real images in this package are qualitative probes without 3D ground truth. Operator benefit, live flight, dense VR readiness, and SOTA image-to-3D ranking remain outside the claim.

Why this is falsifiable novelty. Primitive assemblies, visual hulls, and multiview silhouette consistency are mature [9, 20, 11, 19]. What is new here is the *identification design*: family graph versus equal-budget generic graph isolates the semantic part prior as a spatial knowledge representation for lightweight twin proxies. If the family graph did not help, H1 would fail under the preregistered margin rule. The paper therefore does not sell a denser mesh, a better neural fitter, or a flight-validated system; it sells a measured KR effect under a sealed protocol.

SPPA: language-assisted part decomposition compiled into deterministic 3D primitives

Language can draft the recipe offline; runtime compiles a reviewed versioned recipe plus telemetry.

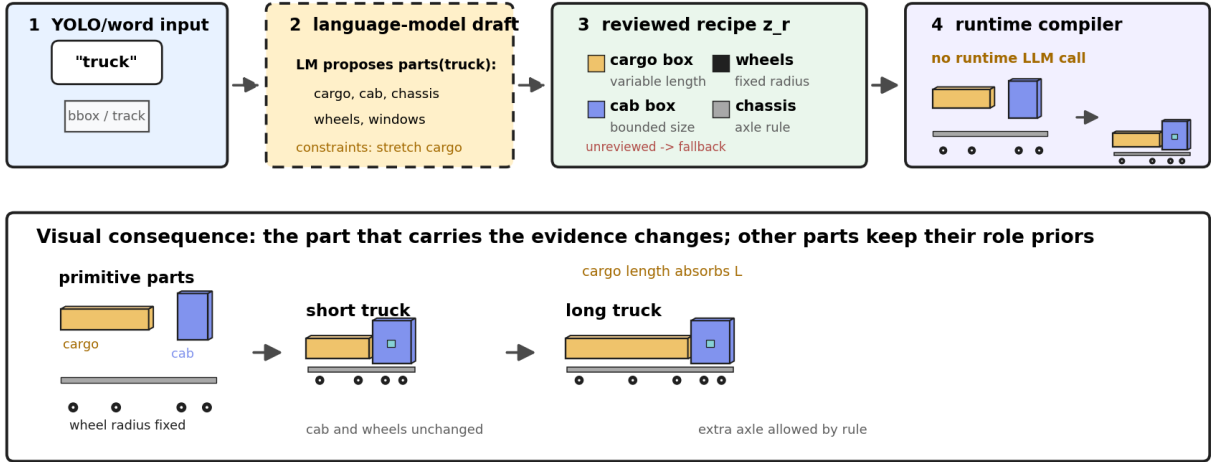


Figure 1: Deployment illustration (not the primary scientific endpoint): semantic label to reviewed part recipe to lightweight primitive proxy for the truck archetype. Offline language/ontology drafting may propose roles, but runtime uses only a reviewed recipe. Short/long variants keep cab and wheel priors while cargo length changes. The confirmatory science is equal-budget family-conditioned MVFit occupancy (Figure 2, Tables 1–4).

Claim boundary: Figure 1 is a deployment illustration of part-parametric proxies, not photoreal reconstruction and not the H1 occupancy result.

Figure 1 shows how reviewed part recipes become twin actors. It is secondary context for the paper’s scientific claim: that family part topology improves occupancy under equal-budget multiview fitting (Section 5).

3 Related Work

We position SPPA-MVFit against three literatures without claiming to replace them.

Neural image/text-to-3D produces dense assets from crops or prompts [16, 10, 12, 8, 18, 22, 21, 23]. SPPA targets sparse semantic telemetry and low-polygon twin actors under shared runtime budgets, not photoreal offline authoring.

Primitive and part-based shape models—including recognition-by-components, superquadrics, learned volumetric primitives, hierarchical semantic parts, and visual hulls—are mature [2, 17, 20, 11, 9, 15, 19]. Our contribution is not a new primitive language, but an equal-budget test of whether a frozen family part graph improves occupancy as a spatial knowledge representation for lightweight proxies.

Digital twins, scene graphs, and mapping layers provide context for object-level visualization [1, 7, 13, 5, 14]. SPPA is a display-actor contract, not a mapping substrate. Operator studies (NASA-TLX, SAGAT) remain future work [6, 3, 4].

4 Family-Conditioned SPPA-MVFit

4.1 Observation contract and shared actor

Every compared method receives the same case identifier, morphology-family token, metric world box, and two 96×96 orthographic binary silhouettes (top and side). No method receives RGB pixels, detector scores, source parameters, ground-truth occupancy, evaluation projections, or real or replay telemetry. The forbidden-input list is frozen in the protocol package.

An actor is an ordered list of eight primitive slots (boxes, cylinders, ellipsoids). Family graphs and a single **generic** graph are stored as static JSON. The only builder used by text-only SPPA, Generic-MVFit, and SPPA-MVFit is

$$A = \text{Build}(g, \theta), \quad \theta = (\log s_x, \log s_y, \log s_z, s_{\text{sec}}, o_{\text{sec}}), \quad (1)$$

with identical bounds for every graph g . Text-only SPPA uses the family graph at the default θ_0 . Generic-MVFit and SPPA-MVFit optimize the same five parameters with the same silhouette disagreement objective and a fixed 31-candidate local search budget; only the graph identity differs (**generic** vs family). This equal-budget construction is the paper’s identification strategy: any occupancy gain is attributed to the family part prior, not to extra parameters, extra views, or a stronger optimizer. Baselines include axis-aligned box, ellipsoid, capsule, billboard, and nonsemantic visual hull [9] from the same masks.

4.2 Preregistered confirmatory protocol

The primary hypothesis compares clean-mask 64-cubed voxel IoU of SPPA-MVFit against Generic-MVFit. Analysis units are 240 source actors (20 per family in each of two strata). The confirmatory estimand is the equal-stratum mean paired difference. H1 passes only if the lower endpoint of a two-sided 95% stratified actor-bootstrap percentile interval (10,000 resamples, seed 77157) exceeds +0.030. Seeds were derived from a NIST Randomness Beacon pulse after a pre-test freeze and a written triple-role protocol release under Amendment 03. Method outputs were sealed and hashed before private ground-truth evaluation. Development results were used only to implement the frozen package and are not confirmatory.

Source actors come from two strata: CSG-ID unions of solids with family topology, and implicit-OOD occupancy from superelliptic extrusions, taper/twist, toroidal sections, and spline-distance tubes. Source and method modules do not import each other. The held-out set is developer-held-out synthetic geometry, not an externally blinded public UAV benchmark.

5 Primary Results

Spatial estimand. The spatial quantity under test is object-level occupancy of a lightweight proxy actor in a shared metric twin frame. Calibrated orthographic top and side silhouettes provide two projected occupancy constraints; 64-cubed voxel IoU is the spatial-quality metric against independent synthetic source occupancy. Family tokens are observed inputs that select a frozen part graph (a spatial prior over role-labeled volumes), not inferred open-set labels. This is a synthetic spatial-proxy evaluation for digital-twin display actors; it is not a cartographic map task and not a measured UAV sensing product.

Table 1 reports the sealed confirmatory decision. Table 2 reports clean-condition method means and wall-clock costs. Table 3 reports secondary paired differences of SPPA-MVFit against the remaining baselines using the same stratified bootstrap interval construction.

Table 1: Preregistered primary endpoint on the held-out synthetic test ($n=240$ actors). Provenance: **synthetic_geometry**. H1 passes only if the stratified bootstrap 95% lower bound exceeds +0.030.

Quantity	Value
Primary mean Δ IoU (SPPA-MVFit – Generic-MVFit)	0.190
Stratified bootstrap 95% CI	[0.181, 0.199]
Prespecified superiority margin	+0.030
H1 decision (lower bound > margin)	PASS
Actors (analysis units)	240
CSG-ID stratum mean Δ	0.209
Implicit-OOD stratum mean Δ	0.172
Resolution sensitivity 48/64/80	PASS

Table 2: Clean-mask method summary on the held-out test ($n=240$). Mean/median are 64-cubed voxel IoU. Timings are single-call wall-clock milliseconds on the benchmark machine and are not Unreal frame times.

Method	Mean IoU	Median IoU	Median ms	p95 ms
SPPA-MVFit	0.557	0.563	9.43	10.60
Generic-MVFit	0.367	0.390	11.34	11.81
SPPA text-only	0.427	0.441	0.77	1.81
Visual hull	0.522	0.567	0.22	0.24
Ellipsoid	0.342	0.292	0.50	0.55
Capsule	0.325	0.260	0.95	1.07
Axis-aligned box	0.248	0.190	0.23	0.27
Billboard	0.173	0.148	0.23	0.26

Table 3: Paired clean-mask mean IoU differences of SPPA-MVFit minus each comparator (stratified actor bootstrap 95% intervals). Visual hull is a high-complexity geometry reference, not a lightweight runtime actor.

Comparator	Mean Δ IoU	95% CI low	95% CI high
Generic-MVFit (H1)	0.190	0.181	0.199
SPPA text-only	0.130	0.116	0.144
Visual hull	0.036	0.027	0.044
Ellipsoid	0.216	0.207	0.224
Capsule	0.233	0.224	0.241
Axis-aligned box	0.310	0.301	0.318
Billboard	0.384	0.375	0.393

H1 passes: mean Δ IoU = 0.190 with 95% CI [0.181, 0.199]. The confirmatory estimand is the equal-stratum paired mean difference; Table 2 is descriptive. The CSG-ID stratum mean difference is 0.209 and the implicit-OOD stratum mean difference is 0.172; neither is non-positive. Table 4 reports all twelve family $\times$ stratum cells from the sealed analysis: every cell is positive, with the smallest gain on compact vehicles under implicit-OOD ($\Delta = 0.043$) and the largest on rider-cycle under CSG-ID ($\Delta = 0.458$). Figure 2 shows illustrative held-out occupancy for three families.

SPPA-MVFit exceeds SPPA text-only (mean $\Delta = 0.130$, CI [0.116, 0.143]) and lightweight geometric baselines. Mean IoU of SPPA-MVFit (0.557) is slightly above the classical nonsemantic visual hull (0.522) [9], but visual hull is denser free occupancy without an eight-part role-labeled actor contract; it is a high-complexity geometry reference, not a defeated runtime competitor. Secondary comparisons use the same stratified bootstrap intervals; null-centered bootstrap p -values with Holm adjustment are recorded in the analysis artifact (Amendment 04). Single-pass clean-call wall time for SPPA-MVFit during the sealed evaluation is median 9.4 ms (p95 10.6 ms) on the benchmark machine; this is not a warm-timing Unreal frame-time claim and is not the protocol H4 multi-call cost design. A resolution-sensitivity check at grids 48/64/80 on the prespecified actor subset passes the protocol threshold. Candidate silhouette renders during fitting used the executed 96×96 observation grid on both arms (Amendment 04 alignment of protocol text with frozen code).

Table 4: Sealed family $\times$ stratum clean-mask mean Δ IoU (SPPA-MVFit – Generic-MVFit). All twelve cells are positive.

Family	CSG-ID Δ	Implicit-OOD Δ	Mean Δ
articulated_vehicle	0.113	0.125	0.119
branching_vertical	0.200	0.210	0.205
compact_vehicle	0.101	0.043	0.072
lattice_tower	0.175	0.163	0.169
quadruped	0.205	0.189	0.197
rider_cycle	0.458	0.299	0.379
Equal-stratum aggregate	0.209	0.172	0.190

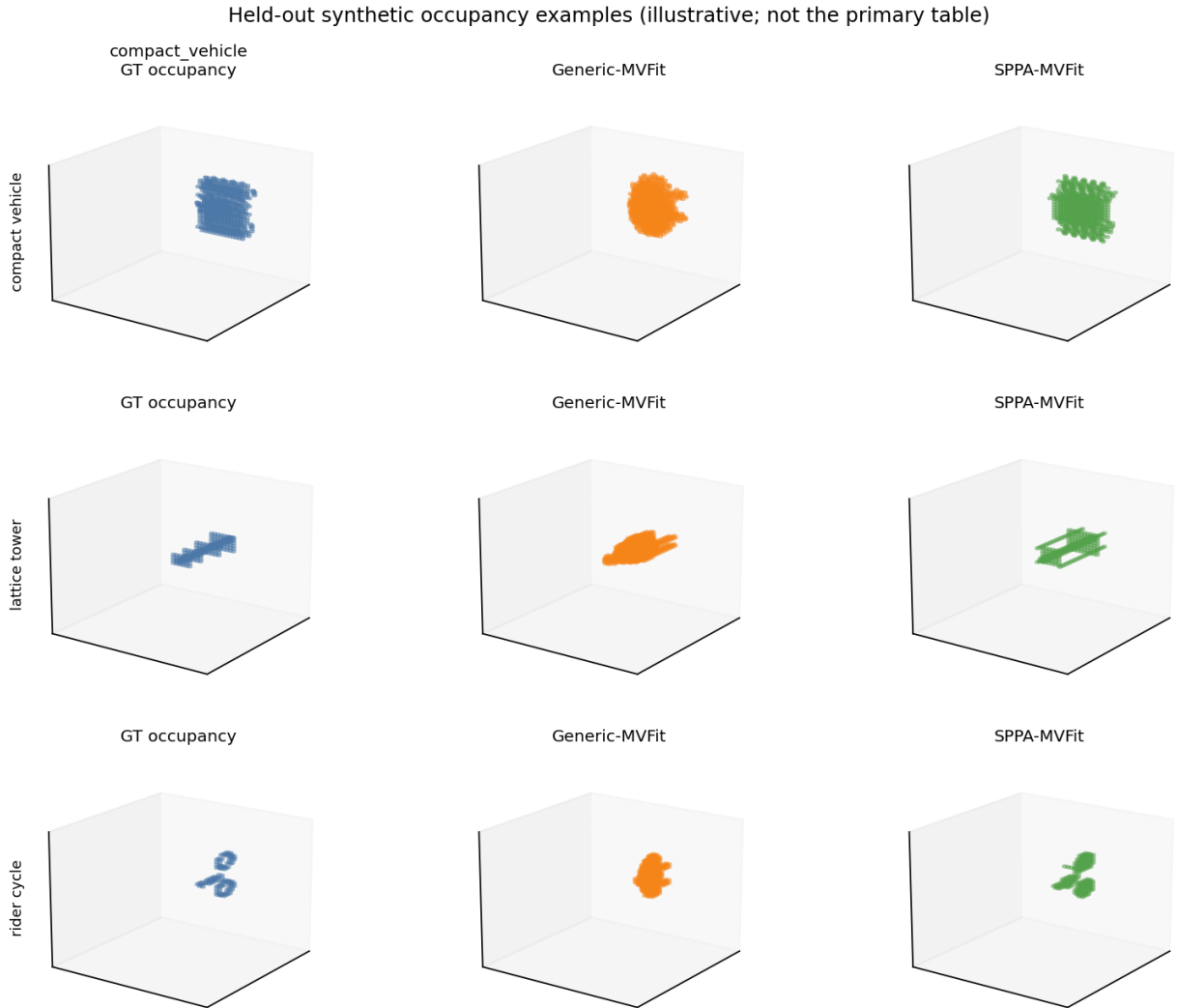


Figure 2: Illustrative synthetic occupancy for three held-out morphology cases (ground-truth source, Generic-MVFit, SPPA-MVFit). The figure is qualitative support for the tables; the confirmatory decision is Table 1.

Claim boundary: these occupancy gains are measured on synthetic calibrated silhouettes with known family tokens. They are not UAV flight measurements, not detector accuracy, and not operator-facing benefit.

6 Deployment Context (Secondary)

Outside H1, production SPPA targets the operational twin use case: sparse semantic evidence to low-polygon role-labeled actors with millisecond local build, optional calibrated footprint dims, mesh LOD (high/balanced/ultra-light), and pose updates without topology rebuild. Table 5 summarizes local operational metrics under balanced LOD (mean about 685 triangles and 0.22 ms build on four probe families; typically 20–60 times fewer triangles than TripoSR-class meshes in the same audit). These metrics support deployment feasibility only; they do not replace the sealed occupancy endpoint and do not claim photoreal image-to-3D ranking or measured flight validity.

Table 5: Operational use-case metrics for SPPA proxies (metric dims, balanced LOD). Not a photoreal image-to-3D leaderboard.

Case	Tris	Parts	Build ms	× lighter vs TripoSR	Use-case score
biker	692	27	0.24	32.5×	0.834
tower	396	17	0.15	61.3×	0.834
tractor	576	28	0.18	63.3×	0.834
tractor_trailer	1076	49	0.33	25.7×	0.834

7 Discussion

The confirmatory result answers the preregistered question: under identical calibrated silhouettes and optimizer budget, a frozen family part graph improves lightweight occupancy over a generic eight-part graph by a large and stable margin. The knowledge representation, not a clever optimizer, is the active ingredient. For a geovisualization / spatial digital-twin venue, the transferable message is that semantic part structure is a spatial prior that can be measured with occupancy IoU on twin-scale actors, without requiring photoreal reconstruction. Secondary gains over text-only SPPA show that silhouette evidence helps when the same family graph is held fixed. Gains over boxes, capsules, and billboards are expected but still quantified under the shared protocol.

Several limits remain binding. The held-out set is synthetic and developer-held-out; positive OOD-stratum differences do not license claims about real UAV masks, detectors, or metric flight geometry. Family tokens are supplied, so the experiment is family-conditioned rather than open-set. Nonsemantic visual hull can approach SPPA-MVFit IoU because it is free to materialize denser occupancy without an eight-part role-labeled actor contract; runtime persistence, updateability, and semantic roles are separate properties. Neural image-to-3D generators remain preferable for offline asset authoring when budget and crops allow, but they are the wrong primary competitor for sparse telemetry under twin runtime constraints. A clean proxy can still look more certain than the underlying evidence; uncertainty encoding is specified, not human-validated.

8 Threats to Validity

First, confirmatory geometry is synthetic and developer-held-out; external public UAV validity is untested. Second, family tokens are observed inputs, not discovered open-set labels. Third, silhouettes are clean orthographic projections plus prespecified corruptions, not real detector masks. Fourth, secondary operational metrics (triangle budgets, local build times) use a different contract from H1 and must not be pooled into one quality score. Fifth, fitter timing is local CPU wall-clock, not Unreal frame time. Sixth, secondary inference uses stratified bootstrap intervals with null-centered p /Holm in the analysis artifact (Amendment 04). Seventh, Amendment 03 was a local protocol-release gate after external review infrastructure failure, not peer review.

9 Claim Boundaries

Supported:

- On the sealed developer-held-out synthetic test, family-conditioned SPPA-MVFit improves clean voxel IoU over equal-budget Generic-MVFit beyond the prespecified +0.030 margin.
- The same shared generator supports text-only and silhouette-fitted modes; silhouette fitting changes θ only.
- SPPA descriptors can be consumed by an optional Unreal backend that coexists with curated assets on the tested packaged/replay paths.

Not supported:

- Real UAV 3D reconstruction accuracy or measured flight telemetry.
- Universal class-agnostic open-set geometry from arbitrary words.
- Operator benefit, safety improvement, or dense VR readiness.
- Image-to-3D SOTA ranking against neural generators.

10 Conclusion

The novelty of this paper is a sealed, equal-budget demonstration that a reviewed semantic-family part graph is a measurable occupancy prior for lightweight multiview proxy fitting. Under a preregistered comparison, SPPA-MVFit improves clean 3D occupancy over Generic-MVFit by 0.190 absolute IoU (95% CI [0.181, 0.199]) on 240 held-out synthetic actors, with positive CSG-ID and implicit-OOD strata and millisecond-scale local cost. The deployment-facing SPPA contract remains a separate engineering layer for compiling imperfect semantic telemetry into updateable Unreal actors when curated assets are missing or mismatched.

The honest next steps are external public data with metric ground truth, detector-derived silhouettes, and—only if claimed—operator studies. Until then, SPPA-MVFit should be judged as synthetic confirmatory evidence for a family-conditioned knowledge representation in lightweight proxy fitting, not as a flight system or universal reconstructor.

Data and Code Availability

The sealed SPPA-MVFit reproducibility package, protocol amendments, NIST-bound seed manifest hashes, and confirmatory summaries are included under `reproducibility/sppa_mvfit/`. Large Unreal artifacts and historical engineering logs remain supporting material and are not required to recompute the primary H1 tables.

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